

How can Game-Based e-Learning Affect Brain Plasticity in Adult L2 Learners?

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### Abstract

This paper uses six studies to make the connection between game-based e-learning and positive structural brain plasticity in adult second language learners. Mårtensson, J., Eriksson, J., Bodammer, N. C., Lindgren, M., Johansson, M., Nyberg, L., & Lövdén, M. (2012) conducted a study to establish that intensive L2 learning creates positive brain plasticity in L2 learners. By first establishing what plasticity is (Spalding, K. L., Bergmann, O., Alkass, K., Bernard, S., Salehpour, M., Huttner, H. B., . . . Frisé, J., 2013) and that it can be affected by intensive L2 learning as well as by video gaming (Anguera, J. A., Boccanfuso, J., Rintoul, J. L., Al-Hashimi, O., Faraji, F., Janowich, J., & Gazzaley, A., 2013), the natural next step was to show how merging video games with a language learning syllabus could lead to cognitive enhancement in adult L2 learners. This paper focuses specifically on how the marriage of video gaming and L2 learning can positively impact vocabulary retention. Secondly, the issue of effective game design, particularly as it pertains to issues of cognitive overload and playability, is also addressed. Hitosugi, C. I., Schmidt, M., & Hayashi, K. (2014) set clear guidelines for what constitutes good game design for L2 learners. A study by Kuwada, K., deHaan, J., & Reed, W. M. (2010) illustrates how playing the video game, Parappa the Rapper 2, led to cognitive overload and inhibited vocabulary retention.

*Keywords:* adult L2 learners, brain plasticity, game-based e-learning, hippocampus

### How can Game-Based e-Learning Affect Brain Plasticity in Adult L2 Learners?

Can spending several hours a week playing a digital language learning game enhance certain cognitive functions in the adult L2 learner? Can these enhanced cognitive functions transfer to untrained tasks beyond the game? Can the language skills acquired to play a 3D video game still be retained long after game play has stopped? Can playing video games actually change the physical structure of the brain, thus leading to positive structural plasticity? The answer to the above questions is a resounding yes.

I believe that ‘the digital age’ has created a new breed of adult L2 learner; one who looks to technology for some or, in some cases, all of their L2 learning needs. I contend that the field of adult ESL/EFL teaching has not taken full advantage of today’s technological resources to create high-quality L2 e-learning experiences that address the specific needs of adult learners. Specifically, I wish to establish that well-designed game-based e-learning can be an ideal tool for inducing brain plasticity in adult language learners thus improving cognitive ability; especially in the area of retention.

### **What is brain plasticity?**

Brain plasticity, also called neuroplasticity or cortical remapping, refers to the physical changes that take place in the human brain as we live through various experiences. More specifically, it refers to the neural pathways that are either created and strengthened from repeated use or weakened from lack of use. For example, an adult studying French for the first time will generate new neural pathways specific to the task of learning French. This is called positive brain plasticity. If this same adult stops practicing the piano to make more time for French, the neural pathways specific to playing the piano will weaken over time. The relationship between positive and negative plasticity is called synaptic pruning, in which some

pathways are strengthened while others are weakened or eventually eliminated (Encyclopaedia Britannica, 2015).

Synaptic pruning enables the brain to run more efficiently by removing un-needed neural pathways in favor of needed neural pathways. At birth, each neuron in the cerebral cortex triggers only 2,500 synapses. However, between birth and age three, the brain develops very quickly, forming excess connections; each neuron now triggering approximately 15,000 synapses. As adults we generate half as many synapses thanks to synaptic pruning (Encyclopaedia Britannica, 2015). Our brain only strengthens the pathways that are required to function in our immediate environment.

### **The hippocampus' role in neurogenesis**

Recent studies in neurogenesis maintain that the adult brain continues to produce a small number of new neurons (approximately 700 / day) in the seahorse shaped part of the brain called the hippocampus, which is responsible for learning, memory, and emotions. While 700 neurons may seem like a drop in the bucket when compared with the 100 billion neurons contained in the human brain, it is believed that these new neurons are what enable brain plasticity. While this rate of neuron production slows down slightly as we age, in most cases it never stops completely (Spalding, 2013).

Additionally, many controllable factors such as aerobic exercise and learning can increase neural production (Lövdén, 2013). Since it is now believed that the brain is able to re-wire itself in response to learning, it would behoove the designers of learning materials to explore the various ways brain plasticity is best facilitated. I contend that multimodal forms of game-based e-learning are an ideal way to develop these neural pathways and make L2 learning more effective for all learners, regardless of age.

### **This is your brain on Dari**

Mårtensson, Eriksson, Bodammer, Lindgren, Johansson, Nyberg, and Lövdén (2012) wanted to investigate whether learning a foreign language resulted in structural changes in the brain, specifically in the areas of cortical thickness and hippocampal volume, and if so, would these changes correspond to language proficiency. The Swedish military has an intensive immersion program for its future interpreters that requires a carefully selected group of candidates to learn a foreign language in 10 months. The 6 women and 8 men from this program had no prior experience learning their respective new languages; Arabic (n=4), Dari (n=8), and Russian (n=2). These 14 participants were brain scanned using an MRI, just before beginning the interpretation program and 3 months after the program started. A control group consisted of 17 medical and cognitive science university students, whose brains were scanned just before starting and just after the completion of one academic semester, approximately 3 months.

The findings showed that, after three months, the cortical thickness in the interpreters' brains was significantly more pronounced than in the control group. Hippocampal volumes were also significantly greater in the interpreter group than in the control group. Within the interpreters' group, differences in the right hippocampus and the superior temporal gyri positively correlated to varying individual levels of language proficiency. Those participants that obtained higher proficiency levels had more malleability in the right hippocampus and superior temporal gyri than participants who had lower proficiency levels.

Such findings indicate that foreign language learning does result in structural plasticity in adult L2 learners. Being able to detect what parts of the brain are affected by L2 learning can help game-based e-learning designers test the efficacy of their games by measuring plasticity in the players. Perhaps incorporating game-based e-learning into the program could have changed

those with low proficiency levels to higher achieving language learners. Can a video game actually change a player's cognitive orientations and how they approach a cognitive task?

### ***Super Mario Bros and Structural Brain Plasticity***

The research team of Kühn, Gleich, Lorenz, Lindenberger, and Gallinat (2014) conducted a longitudinal experiment using a popular 3D gaming platform, Super Mario Bros (SMB), to assess how playing this game affected structural plasticity in the hippocampus and the right dorsolateral prefrontal cortex (DLPFC). Specifically, they studied how players oriented themselves between egocentric (more prevalent in adults) and allocentric navigation in a 3D environment. The top half of the game screen was a third person egocentric perspective of the game environment, always viewing the back of the SMB main character, Mario, and the bottom half of the screen was an allocentric bird's eye view of the game environment. Prior to playing the game, both the SMB and control groups were tested to determine each individual's orientation preference: allocentric, which involves the hippocampus, or egocentric, which involves the caudate.

The study consisted of 48 naïve participants with minimal or no game use six months prior to the experiment. These participants were divided into two groups, those who played at least 30 minutes of SMB every day for the following two months on a portable Nintendo console (n=23) and a control group (n=25) that did not play the game but participated in the testing. The mean age for both groups was approximately 24 years old, with female (n=34) and male (n=14) participants divided fairly equally between the control group and SMB group. Structural changes in the brain were documented through the use of images collected from a head coil. Each participant's perception of the game was documented with questionnaires.

Ultimately, the researchers concluded that a shift from an egocentric orientation to an allocentric orientation in the SMB group resulted in more gray matter production in the right side of both the hippocampus and the DLPFC. The control group exhibited no significant changes in their gray matter volume in these two parts of the brain. Researchers also attributed the gray matter production in both the hippocampus and the DLPFC with the participants' enjoyment in playing the game.

Combining education with game play seems a natural marriage. While the participants were only required to play the game for 30 minutes a day, the average daily rate of play per participant was approximately 50 minutes. Creating a learning tool that learners are motivated to repeatedly come back to should be one of the major goals of e-learning designers. That being said, I would also like to know how the male participants' enjoyment of the game compared with the female participants. As the basic narrative of the game is a brave man going on an adventure to save a helpless princess, the premise of the game may resonate less with the young female players than the young men.

These findings are interesting but this study could have been more rigorous. Including a third study group playing a 2D navigation game might have offered a relevant facet to this study. Finally, all the participants were young adults. How would this test affect the gray matter production in a senior citizen?

### ***NeuroRacer and the Senior Brain***

Anguera, Boccanfuso, Rintoul, Al-Hashimi, Faraji, Janowich, and Gazzaley (2013) conducted a study to test the cognitive control of senior citizens when playing a 3-D video game called *NeuroRacer*. Specifically, the researchers wanted to test how playing this game would affect a healthy senior citizen's ability to multitask. This study started with the premise that our

ability to multitask starts to gradually diminish after our twenties, as our ability to filter out distractions or ‘interference’ also decreases with age. The researchers sought to prove that a multi-tasking video game could elicit positive plasticity in the senior brain and generate lasting effects.

To establish that the cognitive ability to multitask declines as one ages, 174 participants aged 20–79, with approximately 30 participants representing each decade, played two versions of *NeuroRacer*: a single task ‘sign only’ version and a multitask ‘sign and drive’ version. The ‘sign only’ version required the player to respond to a sign only when a green circle appeared on the screen. The ‘sign and drive’ version required the same task as the ‘sign only’ version but the player also had to keep their virtual car on the road using a joystick. As hypothesized, a decline in multitasking abilities positively correlated with increased age, the greatest discrepancy between adjacent decades being those in their twenties (-26.7% cost) compared with those in their thirties (-38.6% cost). The next stage of this study focused on the multitasking abilities of healthy senior citizens. The participants consisted of 46 seniors (ages 65-80), with minimal prior gaming experience, who were divided into three groups: 1. A multitasking group (n=16) that played the ‘sign and drive’ game exclusively; 2. A single tasking group (n=15) that divided their game play between ‘drive only’ and ‘sign only’ versions of *NeuroRacer* in equal measure; and 3. A no contact control group (n=15) that did not play any version of *NeuroRacer*. The multitask and single task groups played *NeuroRacer* for a total of 12 hours over the span of four weeks. Follow-up assessments occurred one month and six months after completion of the four week period, using electroencephalography (EEG) to measure brain activity.

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## Conclusion

The studies linking 3D video gaming and brain plasticity indicate that there are many benefits to be derived from incorporating gaming into an adult L2 learning syllabus. A video game has the potential to strengthen neural pathways and reverse cognitive decline as is evidenced by the studies done using *NeuroRacer* and *Super Mario Bros. EverQuest 2* and *Food Force* show that educational game-based e-learning is not only enjoyable and engaging for the learner but also effective. The video games covered here mostly address the issue of vocabulary retention. While this is certainly an important aspect of L2 learning, *EverQuest 2* also addressed the communicative side of L2 learning. Many schools around the world have started experimenting with the use of discussion boards to foster intercultural competence and L2 writing fluency (Liaw, 2006; Baker, 2013). Games that incorporate this type of interaction can take isolated vocabulary building and put that knowledge to practical communicative use. More studies on how L2 learning could be impacted by these various types of games may change the way learners choose to educate themselves and how teachers choose to utilize or incorporate these resources into their curriculum.

There are numerous studies confirming that DGBLL can be an enjoyable and effective language learning tool either integrated within a traditional school-based curriculum or on its own. The question becomes which game-based tools are best for which students and to develop which skills. It has been suggested that students with a lower L2 proficiency may suffer from cognitive overload if the game play is too complicated. Students with higher proficiency levels may be better equipped to navigate, for example, immersive virtual reality games that have a more complex set of rules. However, good game design targeting specific proficiency levels should address this issue.

### **DGBLL for the future**

Good gaming means different things to different people. One aspect of gaming to consider is the physical component of play. Games like Wii and Xbox require the player to get out of their seat and move their whole body when playing. This has been shown to be beneficial for improving cognitive abilities in the senior population (Hughes, Flatt, Fu, Butters, Chang, and Ganguli, 2014). This active participation may also favor kinesthetic L2 learners who are partial to learning through physical activity. One study touched upon how perceptions of gaming may vary between males and females. Can 2D games be just as effective as 3D games? Are cooperative games more or less conducive to learning than competitive games? Would violent video games enhance or inhibit the cognitive functions related to L2 learning?

The role of certain video games is clearly linked to changes structural brain plasticity. As there is evidence that cognitive enhancements induced by gaming are both long-lasting and transferable, I would conclude that well-designed game-based e-learning has tremendous potential as a highly effective L2 learning tool. As the studies chosen for the topic did not include language learning games designed specifically for adult L2 learners, further research could explore the efficacy of these games, if they exist. If they don't, the findings of this paper could inform future game design.

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